Workshop 1.2: Data Centre as National Infrastructure

Phase 1: Digital Foundations

Workshop Code: WS 1.2
Primary Component: Data Centre Expansion
Duration: 1.5 days
Timing: Month 2, Week 2-3
Phase Theme: "Building Trust & Infrastructure"

Executive Summary

This workshop establishes the strategic vision for Trinidad & Tobago's Tier-4 Government Data Centre as critical national infrastructure that will serve as the foundation for digital transformation. Moving beyond traditional hosting services, the workshop positions the data centre as a catalyst for innovation, a regional digital hub, and an enabler of public-private partnerships. Participants will explore hybrid cloud strategies, green infrastructure approaches, and the data centre's evolution into a technology sandbox supporting AI, ML, blockchain, and IoT initiatives.

Key Outcome: Consensus on data centre expansion strategy, hybrid cloud architecture, and pathway to becoming a regional tech hub.

Workshop Objectives

Primary Objectives

1. **Strategic Positioning:** Establish data centre as foundational infrastructure for e-ID, interoperability, and innovation
2. **Hybrid Cloud Strategy:** Define optimal balance between on-premise Tier-4 infrastructure and cloud services
3. **Green & Inclusive Operations:** Design for environmental sustainability and equitable access
4. **Regional Hub Vision:** Position T&T data centre as Caribbean resource
5. **Innovation Platform:** Plan for tech hub and sandbox environments (AI/ML/blockchain/IoT)

Secondary Objectives

- Build understanding of Tier-4 data centre standards and requirements
- Learn from EU best practices in government data centres
- Establish data sovereignty and security frameworks
- Identify capacity building needs for operations and management
- Begin vendor engagement strategy

Target Participants (25-35 people)

Core Stakeholders (Required)

- **MPAAI Leadership** (4-5)
- Data Centre Program Director
- Chief Information Officer
- Infrastructure leads

- Procurement specialists

- **Current Data Centre Operations** (4-5)

- Facility management
- Technical operations

- Security team

- **Technical Planning Team** (3-4)

- Cloud architects
- Network engineers
- Systems administrators

Extended Stakeholders (Important)

- **Ministry of Public Utilities** (2)
- Energy/power infrastructure

- Sustainability representatives

- **Environmental/Sustainability Experts** (2)

- Green building specialists

- Carbon footprint analysts

- **National-CSIRT** (2-3)

- Physical and cybersecurity

- **Selected MDAs** (3-4)

- Heavy infrastructure users
- Early adopter agencies

Expert Resources

- **EU Data Centre Experts** (2-3)
- Nordic green data centre specialist (Sweden/Denmark/Finland)
- Government cloud strategy expert

- Tier-4 facility design consultant

- **CAF & EU Delegation** (2)

- **Infrastructure Vendors** (2-3)

- By invitation only, for market insights session

Detailed Agenda

DAY 1: Strategic Vision & Infrastructure Requirements

Morning Session (9:00 AM - 12:30 PM)

9:00 - 9:30 | Opening & Strategic Context
- Welcome by MPAAI leadership
- Workshop objectives and overview
- Connection to e-ID (WS 1.1) and upcoming interoperability initiatives
- **Key Message:** Data centre is not just IT infrastructure—it's economic infrastructure

9:30 - 11:00 | The Data Centre as National Asset
- **Presentation:** "Government Data Centres in the Digital Age" (30 min)
- Evolution from server rooms to strategic platforms
- Economic impact: jobs, innovation, digital economy enablement
- International comparisons and benchmarks
- Tier-4 vs. Tier-3 standards explained

- **Case Study:** "Nordic Government Data Centres" (30 min)
- Presented by EU expert
- Sweden's green data centre initiatives
- Denmark's waste heat recovery programs
- Finland's hybrid cloud approach

- Public-private partnership models

- **Panel Discussion:** "T&T Data Centre: Current State & Future Vision" (30 min)

- Current facility capabilities and constraints
- Growth projections based on e-ID and interoperability
- Regional demand potential
- Sustainability imperatives

11:00 - 11:15 | BREAK

11:15 - 12:30 | Hybrid Cloud Strategy
- **Presentation:** "Designing Hybrid Government Cloud" (30 min)
- Cloud-first vs. data sovereignty
- When to use public cloud, private cloud, on-premise
- Multi-cloud strategies
- Cost-benefit analysis framework

- **Interactive Exercise:** "Workload Mapping" (40 min)
- Small groups categorize government workloads:

  - **Group 1:** High-security/sovereign (e-ID, citizen data, defense)
  - **Group 2:** Medium-security (general government services)
  - **Group 3:** Public-facing (websites, open data)
  - **Group 4:** Innovation/sandbox (experimental, dev/test)

For each category, determine:
- On-premise vs. cloud placement
- Redundancy requirements
- Performance needs
- Compliance considerations

- **Report Back** (5 min)

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 5:00 PM)

1:30 - 3:00 | Green Infrastructure & Sustainability
- **Presentation:** "Green Data Centres: Business & Environmental Case" (25 min)
- Energy efficiency metrics (PUE - Power Usage Effectiveness)
- Renewable energy integration
- Cooling innovations (free cooling, waste heat recovery)
- Water conservation

- Circular economy principles
- Carbon neutrality pathways

- **Case Study:** "Sweden's Climate-Neutral Data Centres" (20 min)
- Presented by Nordic expert
- Waste heat used for district heating
- 100% renewable energy

- Lessons learned and cost implications

- **Interactive Workshop:** "Greening the T&T Data Centre" (45 min)

- Breakout groups address:

  - **Group 1:** Renewable energy sourcing (solar, wind potential in T&T)
  - **Group 2:** Cooling optimization (tropical climate considerations)
  - **Group 3:** Waste heat recovery (potential uses in T&T context)
  - **Group 4:** Metrics and reporting (emissions tracking, ESG goals)

Develop concrete recommendations for T&T implementation

- **Group Reports** (10 min)

3:00 - 3:15 | BREAK

3:15 - 4:30 | Inclusion, Access & Regional Dimensions
- **Presentation:** "Data Sovereignty & Equitable Access" (20 min)
- Data sovereignty principles
- Privacy and data protection requirements
- Ensuring equitable access for all MDAs
- Small agency support models
- Gender and inclusion metrics for digital infrastructure

- **Regional Positioning Discussion:** "Caribbean Digital Hub" (30 min)
- Could T&T data centre serve regional needs?
- CARICOM data sovereignty interests
- Revenue potential and economic benefits
- Technical and legal requirements

- Competitive landscape (other regional offerings)

- **Interactive Discussion:** "Inclusion by Design" (25 min)

- How to ensure small MDAs can leverage infrastructure
- Support for capacity-constrained agencies
- Accessibility considerations for all users
- Preventing digital divide within government

4:30 - 5:00 | Day 1 Synthesis & Homework
- Key insights from Day 1
- **Overnight Task:** Participants review technical specifications and vendor information
- Preview of Day 2 (tech hub and implementation)

DAY 2: Tech Hub, Implementation & Next Steps

Morning Session (9:00 AM - 12:30 PM)

9:00 - 9:15 | Day 1 Recap & Day 2 Overview

9:15 - 10:45 | From Data Centre to Tech Hub
- **Presentation:** "Tech Hubs and Innovation Sandboxes" (30 min)
- What is a tech hub vs. traditional data centre?
- Sandbox environments for emerging technologies
- Supporting AI/ML, blockchain, IoT experimentation
- University and startup partnerships
- Economic development potential

- **Case Study:** "Barcelona Supercomputing Centre & Smart City Lab" (20 min)
- Government infrastructure as innovation platform
- Public-private-academic partnerships
- Startup incubation

- Skills development and jobs creation

- **Visioning Exercise:** "T&T Tech Hub 2028" (35 min)

- Small groups envision the data centre in 3 years:
    - What services does it offer?
    - Who are the users (government, academia, private sector)?
    - What innovations has it enabled?
    - What skills/jobs has it created?

- Create vision posters

- **Gallery Walk & Discussion** (20 min)

10:45 - 11:00 | BREAK

11:00 - 12:30 | Technical Requirements & Architecture
- **Presentation:** "Tier-4 Data Centre Technical Requirements" (30 min)
- Tier-4 standards (99.995% uptime)
- Redundancy requirements (N+1, 2N, 2N+1)
- Physical security standards

- Power and cooling infrastructure
- Network connectivity (redundant fiber, submarine cables)
- Disaster recovery and business continuity

- **Technical Workshop:** "Architecture Planning" (50 min)
- Breakout groups (technical focus):
  - **Group 1:** Compute & Storage (servers, storage, virtualization)
  - **Group 2:** Network & Connectivity (internal, internet, regional)
  - **Group 3:** Security (physical, cybersecurity, access control)
  - **Group 4:** Power & Cooling (redundancy, efficiency, backup)

Each group develops requirements document outline

- **Synthesis** (10 min)

12:30 - 1:00 PM | LUNCH

Afternoon Session (1:00 PM - 4:00 PM)

1:00 - 2:00 | Market Insights & Vendor Landscape
- **Structured Vendor Presentations** (45 min)
- 2-3 pre-selected vendors (15 min each)
- Latest technology trends
- Hybrid cloud solutions
- Green infrastructure innovations
- Not sales pitches—market education only

- **Q&A with Vendors** (15 min)
- Clarifying questions only
- Focus on capabilities, not procurement

2:00 - 3:00 | Implementation Planning
- **Presentation:** "From Vision to Reality" (15 min)
- Typical data centre expansion timelines
- Phased implementation approach
- Quick wins vs. long-term investments
- Procurement considerations

- **Workshop:** "Implementation Roadmap" (40 min)
- Breakout groups develop roadmap phases:
  - **Phase 1 (Months 1-6):** Quick wins and planning
  - **Phase 2 (Months 7-12):** Core infrastructure deployment
  - **Phase 3 (Year 2):** Tech hub activation
  - **Phase 4 (Year 3):** Regional expansion

For each phase, identify:
- Key milestones
- Dependencies
- Procurement needs
- Risks

- **Report Back** (5 min)

3:00 - 3:15 | BREAK

3:15 - 4:00 | Closing Session
- **Governance & Operations Discussion** (20 min)
- Data centre governance model
- Operations and maintenance planning
- Capacity building and training needs
- Performance metrics and KPIs

- **Next Steps & Action Items** (15 min)
- Working groups formation
- Follow-up technical assessments
- Procurement planning

- Link to upcoming workshops (WS 3.1, 3.2, 3.3)

- **Workshop Evaluation & Closing Remarks** (10 min)

Key Discussion Questions

Strategic Questions

- What role should the government data centre play in T&T's digital economy?
- Should the data centre serve regional Caribbean needs? Under what model?
- How do we balance data sovereignty with cost-effective cloud services?
- What is the business model for sustainability (cost recovery, revenue generation)?

Technical Questions

- What is the right balance between Tier-4 on-premise and public cloud?
- How do we ensure 99.995% uptime in the T&T context (power, connectivity)?
- What redundancy is truly necessary vs. nice-to-have?
- How do we future-proof for emerging technologies (quantum, edge computing)?

Sustainability Questions

- What is a realistic PUE (Power Usage Effectiveness) target for tropical climate?
- Can we achieve carbon neutrality? By when?
- What renewable energy options are viable in Trinidad & Tobago?
- How do we measure and report on environmental impact?

Implementation Questions

- What can be accomplished in the 12-month project timeframe?
- What requires longer-term investment beyond this project?
- How do we phase implementation to deliver value early?
- What are the critical procurement decisions?

Expected Outputs & Deliverables

Immediate Outputs (During Workshop)

1. **Strategic vision statement** for data centre as national infrastructure
2. **Hybrid cloud workload placement framework** (on-premise vs. cloud decision criteria)
3. **Green infrastructure targets** (PUE, renewable energy %, carbon reduction goals)
4. **Tech hub concept outline** (services, target users, governance)
5. **Technical requirements summary** (Tier-4 specifications, capacity needs)
6. **Phased implementation roadmap** (draft, 3-year view)

Post-Workshop Deliverables (2 weeks)

1. **Workshop Report** (Product 3)
2. Executive summary
3. Strategic positioning and vision
4. Technical findings and recommendations
5. Sustainability framework
6. Implementation roadmap
7. Next steps
8. **Hybrid Cloud Strategy Document** (Appendix)
9. Workload categorization
10. Placement criteria
11. Cloud provider evaluation framework
12. Cost-benefit analysis approach

13. **Green Infrastructure Plan** (Appendix)

14. Sustainability targets and metrics
15. Renewable energy integration strategy
16. Energy efficiency measures

17. Environmental monitoring framework

18. **Tech Hub Concept Note** (Appendix)
    - Vision and objectives
    - Target services and users
    - Governance model
    - Connection to Workshops 3.1 and 3.2

Success Metrics

Participation Metrics

- 100% representation from MPAAI data centre team
- EU expert active participation throughout
- Vendor presentations provide value (rated 4+/5)

Engagement Metrics

- 80%+ active participation in technical workshops
- Consensus on hybrid cloud strategy
- Green infrastructure targets agreed

Satisfaction Metrics

- 4.0/5 average participant satisfaction
- 85%+ found content relevant to their role
- 90%+ understand data centre strategic importance

Outcome Metrics

- Clear technical requirements documented
- Implementation roadmap with timeline
- Sustainability commitments established
- Tech hub vision aligned with WS 3.1/3.2

Connection to Other Workshops

Builds Foundation For:

- **Workshop 3.1:** Tech Hub Launch - Open Source & Standards
- Tech hub concept developed here will be activated in WS 3.1
- **Workshop 3.2:** Emerging Technologies (AI/ML/Blockchain/IoT)
- Infrastructure planned here enables sandbox environments
- **Workshop 3.3:** Private Sector & Third-Party Engagement
- Infrastructure capacity enables third-party services

Inputs From:

- **Workshop 1.1:** e-ID Ecosystem Kickoff
- e-ID infrastructure requirements inform data centre capacity planning
- **Product 1:** Inception Report

Feeds Into:

- **Product 2:** Technical Progress Analysis 1 (end of Q1)
- **Data centre procurement planning** (Month 3-6)
- **Workshop 4.2:** Security, Ethics & Governance (consolidates all infrastructure policies)

Pre-Workshop Preparation

6 Weeks Before

- [ ] Confirm date and book venue
- [ ] Send save-the-date to stakeholders
- [ ] Book EU data centre experts (Nordic green infrastructure specialist)
- [ ] Identify and invite vendor presenters (market education only)

4 Weeks Before

- [ ] Finalize detailed agenda
- [ ] Prepare presentation materials
- [ ] Gather current data centre specifications and capacity data

- [ ] Send formal invitations

2 Weeks Before

- [ ] Send pre-reading materials:
- Current T&T data centre assessment
- Nordic green data centre case studies
- Tier-4 standards overview
- Hybrid cloud strategy frameworks
- [ ] Finalize vendor presentation topics and review slides
- [ ] Confirm catering and logistics

1 Week Before

- [ ] Send logistics reminder
- [ ] Prepare breakout materials
- [ ] Technical setup for presentations
- [ ] Finalize facilitator guide

Materials & Resources Needed

Presentation Materials

- PowerPoint slides for all sessions
- Video tours of leading data centres (if available)
- Infographics on sustainability metrics
- Technical architecture diagrams

Participant Materials

- Workshop agenda and participant guide
- Technical requirements worksheets
- Tier-4 standards reference document
- Green infrastructure metrics guide
- Roadmap planning templates
- Notebook and pen

Technology

- Projector and screen
- Laptop for presentations
- Microphones for panel discussions
- WiFi access

- Flip charts and markers for visioning exercises
- Large poster paper for tech hub vision activity

Documentation

- Attendance tracking
- Photography for documentation
- Note-takers for breakout sessions
- Audio recording (with consent) for complex technical discussions

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Budget Estimate

Category	Item	Estimated Cost
Venue	1.5-day conference room + 4 breakout rooms	$2,500
Catering	Coffee breaks (3), lunches (2)	$2,500
Travel	EU expert flights & accommodation (2 experts)	$18,000
Materials	Printing, supplies, poster materials	$600
Technology	AV equipment rental	$800
Vendor Participation	No cost (vendors cover own travel)	$0
Facilitation	If external facilitator needed	$3,000
Site Visit	Optional: Visit to existing data centre facility	$500
Contingency	10%	$2,790
TOTAL		**$30,690**

Risk Mitigation

Risk	Mitigation
Vendor presentations become sales pitches	Pre-approve all content; strict time limits; frame as "market education"

Risk	Mitigation
Technical complexity overwhelms non-technical participants	Balance technical depth with strategic content; use analogies; provide glossary
Unrealistic sustainability targets	Ground in EU case studies with similar climate/ context; expert guidance on feasibility
Scope creep (trying to solve everything)	Clear workshop scope; parking lot for out-of-scope items; phased roadmap
Lack of consensus on cloud strategy	Present as framework, not final decision; emphasize flexibility and evolution

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Gender & Inclusion Considerations

Representation

- Include women engineers and technical leads in presentations
- Invite sustainability experts who can speak to social equity dimensions
- Ensure environmental justice perspectives represented

Content

- Discuss how data centre location affects communities (jobs, environment)
- Address digital inclusion through infrastructure access
- Consider gender dimensions of tech jobs and skills development

Process

- Technical workshops accessible to various skill levels
- Multiple ways to contribute (technical, strategic, policy)
- Acknowledge and value diverse expertise

Follow-up Actions (Post-Workshop)

Week 1

- [ ] Send thank you emails
- [ ] Distribute photos and key visuals
- [ ] Begin drafting workshop report

- [ ] Compile technical requirements document

Week 2

- [ ] Finalize and circulate draft workshop report
- [ ] Share with procurement team for vendor planning
- [ ] Schedule technical working group meetings
- [ ] Update project timeline based on roadmap

Week 3-4

- [ ] Incorporate feedback and publish final report
- [ ] Submit to CAF and EU as Product 3
- [ ] Begin detailed technical specifications development
- [ ] Plan for Workshops 3.1, 3.2, 3.3 (tech hub activation)

Month 3-6

- [ ] Procurement planning and vendor engagement
- [ ] Detailed hybrid cloud architecture design
- [ ] Sustainability assessment and renewable energy sourcing
- [ ] Site assessments and facility planning

Links to TOR Requirements

This workshop directly addresses:

TOR Section 5, Objective C: Data Centre Expansion Feasibility Actions
- **Action a:** "Ensure strategies are designed for greener, inclusive operations, including metrics on emissions reduction and access to identity rights."
- **Action b:** "Procure and deploy a hybrid cloud solution supporting regional data sovereignty and cross-border services."
- **Action c:** "Support the conduct of workshops (approx. 5) to develop the data centre as a tech hub with sandboxes for open-source, Artificial Intelligence (AI), Machine Learning (ML), blockchain, and Internet of Things (IoT)." [This is the first of the series]
- **Action f:** "Ensure that all data management activities take into account international best practices, standards, and digital solutions, to maximize security, accessibility, and efficiency."

TOR Section 6, Services to be Carried Out
- **Service 5:** "Develop strategies to engage Caribbean and international stakeholders, particularly from smaller countries that represent international best practice..."
- **Service 6:** "Actively seek and evaluate international solutions, technologies, and best practices that can be incorporated into the various project components."

TOR Section 7, Products
- **Product 3:** Workshop Reports

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Next Review: 2 weeks before workshop execution